WELLSLING · USER REQUIREMENTS

WellSling user requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First issue (kit 1.9.1 upgrade), drawn from the users, stakeholders, constraints and settled questions in WSL-PRB-001

WellSling user requirements

What the people who would use WellSling need, written from their side. The product requirements that meet these needs are in WSL-PRS-001 (`03-requirements.md`) and the links are in the traceability matrix WSL-RTM-001 (`03b-traceability.md`). Sources are the users and stakeholders in WSL-PRB-001, Tables 1 to 3 (users, users and needs, stakeholders). These are proposals until a co-design partner reviews them. Method: STANDARDS section 22.

> **Safety:** The first need is that nobody goes down a well. WellSling is an open engineering reference, not certified lifting or rescue equipment.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Safety	Nobody goes down	We can get the animal out without anyone climbing into the well at any stage, because the air at the bottom can kill	High	Farmers and livestock owners; families of rescuers (PRB Tables 2 and 3)
UR-002	Performance	A grown cow or buffalo	It lifts a grown cow or buffalo, up to about 600 kg, and a calf	High	Farmers and livestock owners (PRB Table 2)
UR-003	Performance	Deep wells	It works when the animal is as far down as the deepest open wells in our area, about 15 m	High	Village volunteers; animal rescue organisations (PRB Table 2)
UR-004	Deployment and assembly	Wide wells, weak rims	It stands over open wells up to about 3 m across without loading a crumbling or wet rim	High	Village volunteers; rural fire crews (PRB Table 2)
UR-005	Deployment and assembly	Quick to set up	Three or four of us can have it ready to lift within about 20 minutes of reaching the well	High	Village volunteers and dairy cooperative staff (PRB Table 2)
UR-006	Usability	Short briefing	Villagers who have never used it can work it safely after a short briefing at the well	High	Village volunteers; dairy cooperatives (PRB Tables 2 and 3)
UR-007	Safety	Stop and hold	The animal stops and stays where it is whenever we let go	High	Farmers; rescue organisations (PRB Table 2)
UR-008	Safety	Gentle on the animal	The lift spreads the animal's weight so it is not injured or crushed while it hangs	High	Animal rescue organisations; veterinarians (PRB Tables 2 and 3)
UR-009	Safety	Land without swinging	The animal is set down on firm footing without being swung over the rim	High	Rescue organisations; rural fire crews (PRB Table 2)
UR-010	Usability	Human power only	It works by hand, with no power on site, at an effort ordinary people can keep up in turns	High	Farmers; village volunteers (PRB Table 2 and operating environment)
UR-011	Safety	Frame does not tip	The frame stays standing when the animal swings or kicks, or a guide line is pulled	High	Rural fire crews; rescue organisations (PRB Table 2)
UR-012	Safety	Nobody falls in	People working at the rim are kept from falling into the well	High	Village volunteers; families (PRB Tables 2 and 3)
UR-013	Logistics	Two-person pieces	Every piece can be carried by two people across a field; the kit fits a small trailer or pickup	Medium	Village volunteers; rescue vans (PRB Table 2)
UR-014	Cost	Affordable to share	A village, cooperative or rescue group can afford one kit and share it	Medium	Dairy cooperatives; NGOs (PRB Table 3)
UR-015	Environment and durability	Lives outdoors	It survives years of storage in a shed or yard, monsoon rain, mud and dung	Medium	Dairy cooperatives (PRB operating environment)

ID	Category	Title	Need	Priority	Source
UR-016	Maintenance	Local repair	A village welder or fabricator can make and repair the steel and timber parts	Medium	Fabricators and workshops (PRB Table 3)
UR-017	Regulatory	Free to copy	The design is open and stays clear of others' patents, so anyone may build it	Low	Open-hardware community; Amish (PRB constraints and design-arounds)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.